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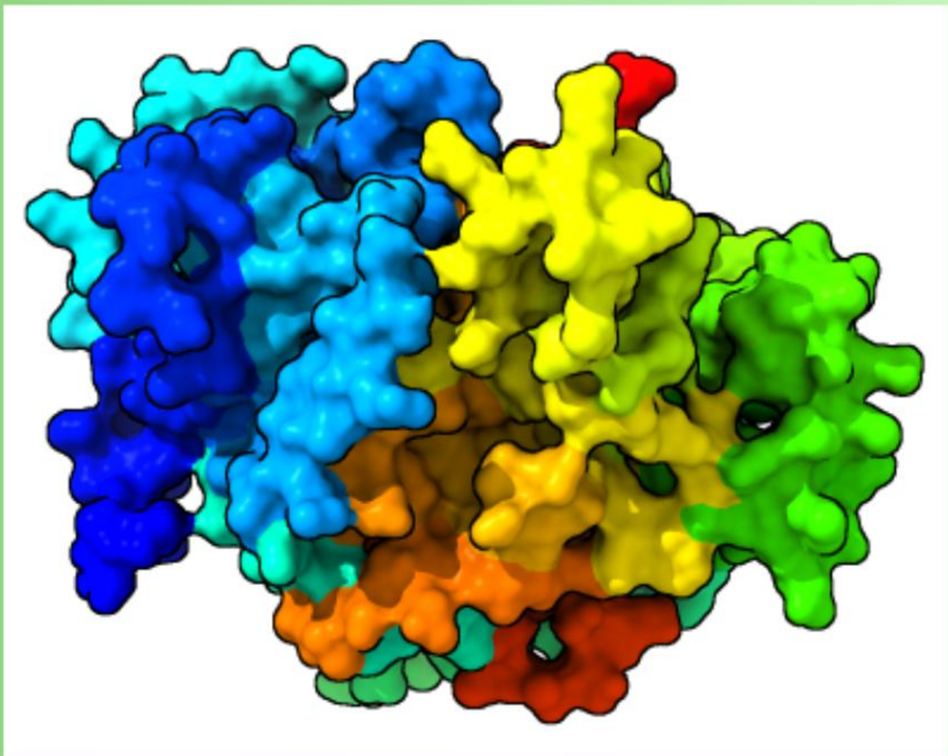
GCN-based Functional Prediction of Uncharacterized Proteins



PERE PALMER SÁNCHEZ-MONGE

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Tutor: Eloi Parladé Molist

Abstract

Protein function prediction remains a major challenge in the postgenomic era because the number of sequenced proteins far exceeds the number experimentally characterized. In this work, a graph convolutional network-based and bioinformatics-driven approach was applied to infer the functions of 26 uncharacterized proteins from *Haloquadratum walsbyi*, an archaeon adapted to hypersaline environments. The proteins were selected from a broader set of hypothetical proteins based on sequence length, since longer proteins provide more informative signals for functional inference.

To increase confidence in the annotations, multiple complementary strategies were combined: I-TASSER for structure-based functional prediction, BLAST searches against related and broader archaeal genomes, COG classification, and KEGG Orthology assignment. The resulting annotations suggested a diverse set of functions, including nucleotide binding, transport, transferase activity, helicase activity, redox metabolism, and lipid-related processes. Several proteins were restricted to *Haloquadratum* or *Haloferaceae* lineages, while others matched known archaeal enzyme families, indicating a mixture of lineage-specific and conserved biological roles.

The analysis also highlighted proteins with potential participation in shared metabolic pathways and a possible protein complex involved in fatty acid metabolism. In addition, several proteins were associated with domains of unknown function, underscoring the persistent difficulty of assigning roles to highly conserved yet poorly characterized protein families. Overall, the results support the use of graph-based and integrative computational methods as effective tools for protein annotation in understudied genomes, while also showing the need for experimental validation to confirm the proposed functions.

Keywords

protein function prediction, uncharacterized protein, GCN, DUF, protein complex

Introduction

The central dogma of molecular biology and its subsequent corrections and revisions is the base of how scientists understand life.[1,2] At its most basic it can be interpreted as nucleotide sequences that codify for a certain amino acid sequence that folds itself in order to minimize entropy, this sequence and structure give each protein their function.[3,4]

Protein function prediction is one of the most important challenges of the postgenomic era, mostly because of the exponential increase in available sequenced proteins, and a significantly slower experimental characterization.[5] This imbalance has caused a significant **annotation inequality,** where a lot of resources and time are dedicated to very few proteins, already well studied, leaving thousands of “hypothetical proteins” with a known or predicted structure but without an assigned function, as well as “understudied proteins”.[6,7]

With the knowledge that 95% of life science publications involving proteins focus on the same group of 5000 *Homo sapiens* proteins, there have been efforts to reduce this knowledge gap, such as the “Understudied Proteins Initiative”.[5,8]

These initiatives use various methodologies to assess the potential function of proteins, from the traditional experimental analyses to advanced computational methods. MALDI mass spectrometry is one of the premier protein identification methods and is extensively used on clinical and environmental samples.[9] On the other hand, bioinformatics permits the analysis of whole protein families and the identification of candidates with promising biochemical properties by predicting their stability and cellular location.[6,10]

A problem arises with moonlighting proteins, which participate in more than one chemical reaction, or even in different metabolic pathways.[11] Another issue is convergent evolution, where non-homologous proteins develop similar functions, such as chymotrypsin and subtilisin, proteinases that share a Ser-His-Asp active site.[6] In order to standardize the functional characterization different systems have been developed, such as the **Enzyme Commission numbers (EC)** for enzymes, and the **Gene Ontology (GO)** for proteins in general, that structures function in three different subontologies: Molecular Function (MF), Biological Process (BP) and Cell Component (CC).[12,13]

Automatic Function Prediction

Computationally, protein functional prediction is what’s called a hierarchical, multi-modal large-scale problem. This results from the fact that the GO system is hierarchical (every term except for “biological function” has at least one “parent term” and can have any number of “child terms”), proteins have multiple GO terms assigned and the GO database having more than 40.000 terms.[12,14]

The **Computational Assessment of Function Annotation (CAFA) standards** were developed as a tool to grade systematically and objectively the different Automatic Function Prediction systems (AFP) developed.[15–17] The CAFA challenge was a time-delayed assessment, where organizers published a set of target protein sequences without any previous experimental functional annotation then, participants sent their computational predictions for these proteins, after a period of time the organizers collect the new experimental annotations that have accumulated for the target proteins, finally this new set of annotations (the "benchmark set") is used as a ground truth to assess the accuracy of the predictions sent.[15]

Three more CAFA challenges have taken place, showing significant growth in prediction quality over the last decade. This improvement was attributed to the growth of experimental annotations and the advancement in the prediction methods themselves. It was also found that most methods have their **best performance for MF**, while BP's performance is lower, and both CC's and the Human Phenotype Ontology's (HPO) annotations are even more difficult to predict.[16–18]

In order to grade each model's performance CAFA uses two main perspectives: the ability of a method to predict **all terms associated with a given protein** and its ability to identify **all proteins associated with a specific functional term**. Both perspectives can offer different interpretations of the performance of a method, especially for CC and HPO ontologies.[15–17] The best-performing methods use a wide range of approaches and data sources (sequence alignment, protein-protein interactions, sequence properties, machine learning), suggesting that **assembly methods that integrate independent predictions could offer substantial improvements**.[19,20]

AFP has gone through several methodological stages, from simple transfer by homology to complex deep learning models. Early computational methods, and still used as a baseline, are based on sequence similarity annotation transfer. The most common tool is BLAST[21] which assigns to an unknown protein the functions of the most similar protein found in a database of already characterized proteins, although it is estimated that **only about 33% of proteins with unknown function have detectable homology with well-characterized proteins**, which severely limits the applicability of this approach.[22]

Faced with the limitations of homology, methods based on **machine learning** emerged. These models use a set of proteins with known functions to learn how to map biologically relevant features to functional labels. The most common algorithms used are **Support Vector Machines (SVMs)**, Random Forests (Random Forests) and k-Nearest Neighbours (kNNs).[22–24] A key step in these methods is the identification and extraction of numerical features (character vectors) that represent proteins. One of the premier examples is GOstruct, a model based on structured SVMs that maps directly from the hierarchically consistent input tags, eliminating the need for multiple binary classifications.[25,26]

Feature	Description	Examples
Sequence-based	Directly derived from the amino acid sequence.	Amino acid composition (AAC), Dipeptide Composition (DPC), PseAAC, sequence motifs, PSSM.
Physical and chemical properties	Based on the physical and chemical properties of amino acids.	Molecular weight, isoelectric point, hydrophobicity (GRAVY), polarity, charge.
PPI Networks	Based on the physical interactions between proteins.	Neighbour connectivity, average shortest path.
Text-mined	Extracted from existing literature. Natural Language Processing (NLP).	Term Frequency (TF-IDF), co-occurrence if protein names and GO terms.
Existing Annotations	Using previous annotations to predict new ones.	Orthologs, protein domains

Table 1. A list of features from which information relevant to AFP can be obtained and are used in the development of new prediction models.[16,19,20,22,24,26]

Deep Learning

Modern Deep Learning represents a paradigm shift from manual feature engineering to the automatic extraction of hierarchical patterns from raw data.[27,28] These networks consist of multiple layers of nonlinear processing units, where **each successive layer uses the output of the previous one to form increasingly abstract representations**. The proliferation of these models was driven by the introduction of the costly layer-wise pre-training, the availability of large-scale datasets, and the development of high-performance hardware. While early neural networks like the multi-layer perceptron (MLP) faced issues such as gradient diffusion and overfitting, modern architectures leverage specific structural priors in data to improve performance.[27]

Convolutional Neural Networks (CNNs) have become the state-of-the-art for processing **grid-like, Euclidean data** such as images and videos.[27–29] Their effectiveness stems from the use of a convolution operator that slides fixed-size trainable filters across pixels to combine local information.[29] This architecture exploits the "stationarity" and "compositionality" of images, allowing the network to capture low-level features like edges in initial layers and complex objects in deeper layers.[27,29] Models like AlexNet, VGGNet, and ResNet demonstrated the power of this approach by achieving record-breaking results in large-scale visual recognition competitions.[27,28] However, CNNs are inherently limited by their reliance on regular grids, making them unsuitable for data with irregular shapes and sizes.[28,29]

On the other side, Graph Convolutional Networks (GCNs) extend Deep Learning to **non-Euclidean structures** to handle irregular structured data; such as molecules, social networks, and protein-interaction networks. Unlike traditional CNNs, GCNs operate on non-Euclidean data, where a natural spatial order doesn't exist, and the number of neighbours for each node can vary.[28,29] This difference between CNNs and GCNs is illustrated on **Figure 1**, obtained from Liang *et al.* 2022.[30] The core mechanism of a GCN is to **update a node's representation by aggregating information from its neighbourhood in a convolutional fashion**.[29,31] This allows the model to capture the intrinsic structural relations among data points that would be lost if analysed in isolation.[29]

GCNs are broadly categorized into spectral- and spatial-based models. Spectral-based models define convolutions in the frequency domain using the graph Fourier transform and the eigenfunctions of the graph Laplacian matrix.[32] Early models faced high computational costs, leading to the development of simplified variants that approximate first-order neighbourhoods.[29] As a contrast, **spatial-based models** define convolutions directly in the vertex domain by aggregating representations from immediate neighbours, a process often described as "message passing". Spatial GCNs are generally more flexible and efficient for large-scale, dynamic graphs.[28,29]

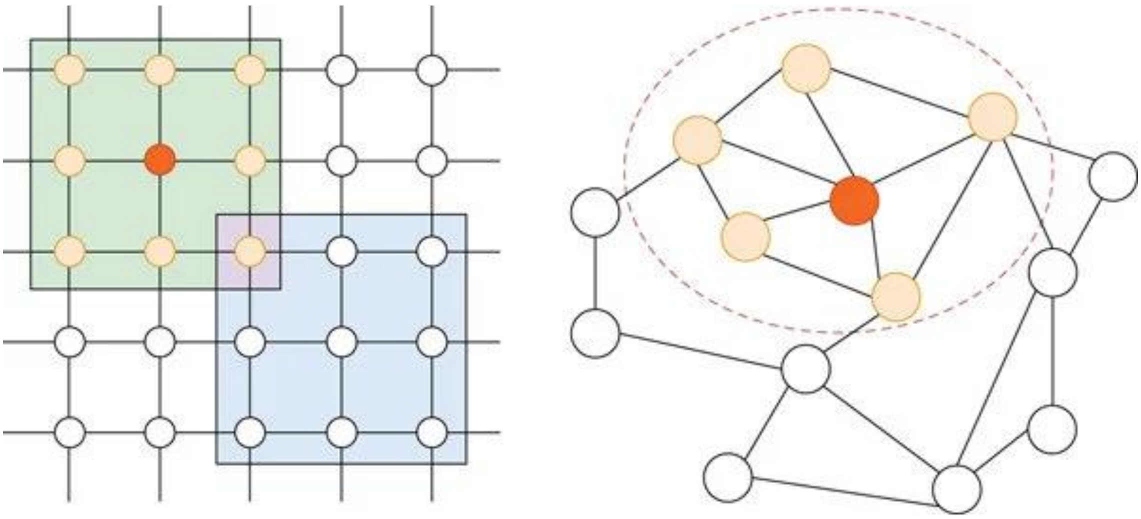

Figure 1. Comparison between an Euclidean graph (left) and a non-Euclidean one (right).[30]

In bioinformatics, GCNs have opened a new paradigm for drug discovery and molecular informatics. Because molecules are naturally represented as graphs (where atoms are nodes and bonds are edges), **GCNs can automatically extract structural information more effectively than handcrafted molecular fingerprints**.[28] Modern frameworks like MVGCN integrate multiple "views" of data, such as drug similarity and disease semantic networks, helping to predict unobserved links in biomedical bipartite networks.[31] These models are actively used for Drug-Target Interaction (DTI) prediction, miRNA-disease association identification, and de novo molecular design.[28,33] Despite their success, some challenges remain, such as "oversmoothing", where nodes become indistinguishable in highly dense graphs, and the need for better interpretability in healthcare applications.[28,29,31]

Model	Main Methodology	Key Focus
BLAST	Local sequence alignment	Baseline for homology-based function assignment.
GOstruct 2.0	Modified structured SVM	Explicit modelling of incomplete annotations to predict new functions in already annotated proteins.
DeepGO	Deep Learning (CNN and Graph Embeddings)	Integrates sequence and PPI, with an architecture similar to GO terms.
MTDNN	Multi-Task Deep Learning	Each GO term prediction is approached as an individual task, although related in order to use the functional dependencies.
Mishra *et al.*'s model	Deep Neural Network ensemble	Specialized focus on hypothetical proteins from pathogens, using a massive feature ensemble.
I-TASSER	Deep Learning integrated pipeline	Generates a structural prediction, gets binding sites, EC and GO from databases, and the function is inferred from protein function databases based on the information accumulated.
I-TASSER-MTD	Deep Learning integrated pipeline	Structural and functional prediction for multi-domain proteins, similar pipeline to I-TASSER.
COFACTOR	Based on sequence, structure and PPI	Tool for the functional annotation integrated into the I-TASSER suite.
GOLabeler	Ensemble (BLAST-kNN, logistic regression)	Based on "Learning To Rank", high performance on the CAFA challenges.

Table 2. Brief comparison of some of the different models that predict protein function.[14,19,21,22,26,34–37]

Specific challenges

More than **two-thirds of prokaryotic proteins and four-fifths of eukaryotes contain multiple domains**. As structural and functional prediction of multi-domain proteins is particularly challenging, due to the degrees of freedom in the orientation of domains, models such as I-TASSER-MTD appeared as platform specifically designed for multi-domain proteins.[35] It includes sequence-based domain analysis to predict domain boundaries, models the structure of each domain separately using D-I-TASSER, which integrates deep learning spatial constraints (DeepPotential).[38] It also couples individual domain models into a complete chain structure using analogue templates and Monte Carlo simulations. A feature that it shares with the standard I-TASSER model is the use of the COFACTOR tool to predict GO terms, EC numbers and ligand binding sites at both the individual domain and full string level.

Existing functional annotations are a reflection of the global knowledge at a given time and, by nature, are incomplete. AFP methods can be divided into two subtasks: making predictions about previously unnotified proteins (NPs); and predicting new functions for already annotated proteins (NAs). The **NA task has been observed to be particularly challenging** for many models, and as such, GOstruct 2.0 was developed to directly address this challenge.[26] It recognizes that the existing annotations of a protein are incomplete, and its main innovation is a modification of the original structured SVM formulation in which a "label extension" is defined as a set of predictions consistent with the existing annotations while including additional or more specific terms. This reduces the penalty for margin violations when the second-best candidate prediction is an extension of the known label. This allows the model to be more flexible and better represent the process of accumulation of annotations over time, where a protein acquires increasingly specific functions.[26]

Experimental results in human data showed that GOstruct 2.0 has significant improvements in the NA task, especially on the Biological Process subontology.[25,26]

AFP models are particularly suited for the study of organisms where not all proteins are characterized, mostly due to the barebones functional analysis carried out when the organism is first isolated and studied. In most cases, the experimental studies are focused just on 1 or 2 proteins, and the rest are given a quick algorithmic prediction of the CDS function and not followed up. One of these organisms is ***Haloquadratum walsbyi***, an haloarchaeon of the *Haloferaceae* family.[39]

Automatic Function Prediction has evolved into a sophisticated field of machine learning, deep learning models have shown considerable success, overcoming traditional approaches thanks to their ability to integrate multiple data sources[34,35] and learn representations of complex characteristics automatically. **Progress is evident and constant**.[17]

However, there are still significant challenges, performance in subontologies such as the Biological Process and the Cell Component remains lower than that of Molecular Function. In addition, the data imbalance is a persistent problem.[16,24]

Future directions of the field will include greater data integration, ensemble development, an improved interpretability of models, and organism-specific approaches. Ultimately, the goal still is to reduce the knowledge gap between sequence and function, which will help to elucidate the mechanisms of diseases and to advance in the design of new proteins and therapies.[17,22,35,37,38,40]

Initial hypothesis and objectives

Main hypothesis: given the state of GCN technology, functional prediction of a series of proteins is accurate enough to assign a putative function.

General objectives

To obtain new information from an understudied genome.

Specific objectives

1. To identify protein function using GCN-based bioinformatics tools.
2. To give context to this information by comparing it to various databases.

Materials and methods

The necessary background information was retrieved from academic search tools such as Google Scholar[41] and PubMed, using queries such as “Protein function identification”, “Protein structural homology” and “Machine learning AND functional proteomics”. The candidate genome was found via search in the NCBI Datasets online database, and its hypothetical proteins sequence and predicted structure were obtained from UniProtKB.[42]

The sequences retrieved were analysed using the I-TASSER online suite,[34] and the “Predicted Functions” were obtained.

In order to give the predicted functions some context a search was carried out in two faces: “Related Organisms”, where a BLASTn and BLASTp search[21] will be carried out against all *Haloquadratum* genomes in the NCBI database, hoping to find orthologous proteins with an assigned function, helping confirm the prediction. And “Similar Organisms”, where a similar BLAST search will be carried out against the whole NCBI database. This search will try to find homologous proteins in unrelated organisms in similar ecological niches.

More information was obtained when the protein sequences and GO annotations were used to predict their Clusters of Orthologous Genes (COG) classification.[43,44] The COG clusters were identified thanks to a bash script developed by Claude Sonnet 4.6.[45]

Finally, in an effort to integrate the newly acquired knowledge into known pathways, KEGG Orthology (KO) numbers were assigned to each protein using the four annotation tools provided by the KEGG web platform (BlastKOALA[46], GhostKOALA[46], KofamKOALA[47] and DeepKOALA[48]).

Results

The *Haloquadratum walsbyi* type strain (DSM16790) has 2558 proteins identified on UniProtKB, with 419 named “Uncharacterized protein”.[42] As can be seen in **Figure 2**, these proteins are significantly more concentrated in the 1 to 200 amino acids range (297/419, 71%) than the whole proteome (963/2558, 38%). Because of the impossibility of analysing 419 proteins for this project, only the 26 proteins longer than 400 amino acids will be studied. The criteria for longer chains was chosen because shorter chains have less total information from which function can be inferred.

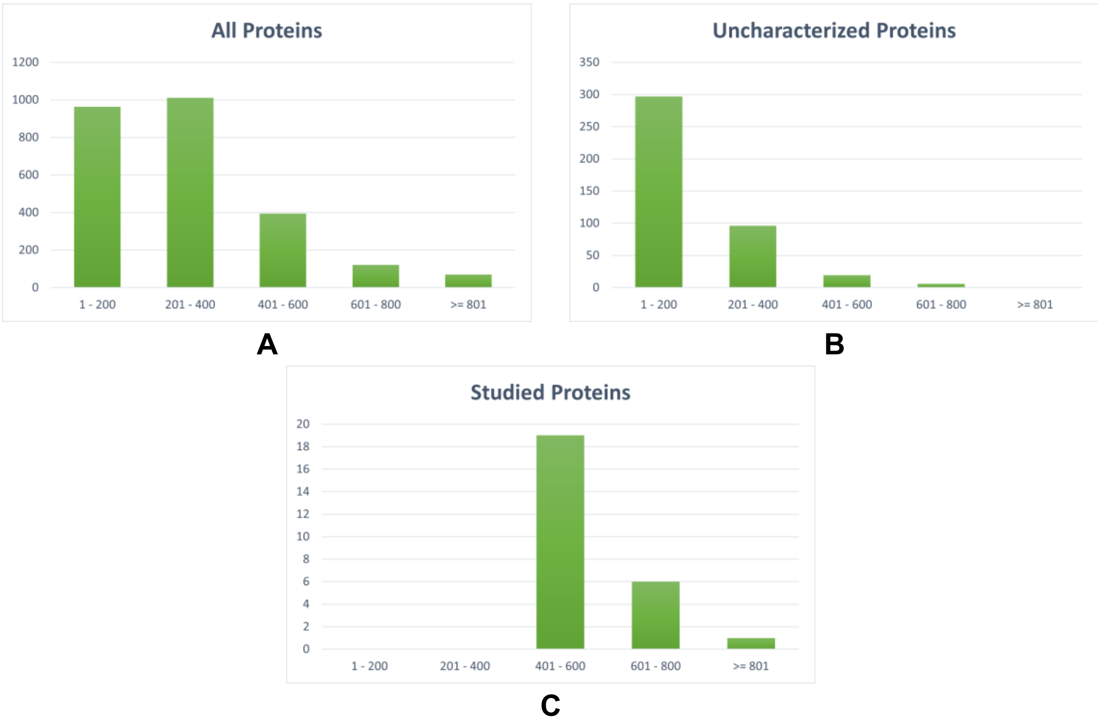

Figure 2. Distributions of protein chain length in the whole proteome (**A**), the uncharacterized proteins (**B**), and the studied proteins (**C**).

Studied Proteins

As shown in the first sheet of **Supplementary Table ST1**, of the 26 proteins studied, 10 contain between 400 and 500 amino acids, 9 contain between 500 and 600 and 8 contain more than 600 amino acids. The UniProtKB accession codes of the proteins are: J7RFT8, Q18E01, Q18EI8, Q18F64, Q18F66, Q18F80, Q18FA8, Q18FD4, Q18FD7, Q18FM1, Q18FW5, Q18GD8, Q18GI1, Q18GI9, Q18H44, Q18H89, Q18HM5, Q18HZ6, Q18IF7, Q18IF9, Q18IK6, Q18IW2, Q18IW6, Q18JC6, Q18JC9 and Q18JT2.[42]

I-TASSER Results

Protein	Predicted Function
J7RFT8	GTPase
Q18EI8	Xanthine phosphoribosyl transferase
Q18F64	Methionine-tRNA ligase
Q18FA8	Acetyl- or Malonyl-transferase
Q18FD7	Adhesion to substrate
Q18GD8	Na+ Transmembrane transporter
Q18GI1	Acyltransferase, Carboxylic acid synthesis subunit
Q18H44	Acyltransferase, Carboxylic acid synthesis subunit
Q18HM5	1,4-alpha-oligoglucan phosphorylase activity
Q18HZ6	Amino acid or nucleobase transmembrane transport
Q18IF7	DNA helicase activity, RNA biogenesis/DNA repair
Q18IW2	Acyltransferase, Carboxylic acid synthesis subunit
Q18JC6	Acyltransferase, Carboxylic acid synthesis subunit
Q18JC9	Beta-galactosidase activity

Table 3. Summary of some of the results obtained from the I-TASSER model.

As can be seen on **Table 3**, the functions of the studied proteins are varied and range from transmembrane transporters to helicases and transferases of various functional groups, such as acyl and phosphate.

These "Predicted Functions" are obtained from the GO codes and internal scores that comprise the I-TASSER output,[34] where not all the proteins obtained a valid predicted function. The complete list of GO annotations[12] and the scores that determine their validity can be found on the second sheet of **Supplementary Table ST1**.

The proteins that interact with nucleotides are J7RFT8, Q18EI8, Q18F64, Q18IF7 and possibly Q18HZ6. J7RFT8 is predicted to be a GTPase, suggesting a nucleotide-binding role, although the exact biological process remains unclear. Q18EI8 is predicted as a xanthine phosphoribosyl transferase, Q18F64 is annotated as a methionine-tRNA ligase, Q18IF7 is predicted to be a DNA helicase, although its exact BP can't be specified and Q18HZ6 is a transmembrane transporter, but can't be classified into amino acid transport or nucleobase transport.

Q18FA8 is predicted as an acetyl- or malonyl-transferase. Q18FD7 is annotated as an adhesin homologue, Q18GD8 is predicted to function as a Na+ transmembrane transporter, Q18HM5 is predicted as a 1,4-alpha-oligoglucan phosphorylase activity protein, and Q18JC9 is predicted to have beta-galactosidase activity.

Finally, Q18H44, Q18GI1, Q18IW2 and Q18JC6 are annotated as acyltransferases that take part in a carboxylic acid synthesis protein complex, and are part of the broader fatty acid metabolism BP.

BLASTn and BLASTp

Protein	Taxonomy	Comments
J7RFT8	Only *Haloquadratum*	DR2241 family protein
Q18E01	Only *Haloquadratum*	DUF7286 family protein
Q18F64	Only *Haloquadratum*	DUF8115 family protein
Q18FA8	*Haloferaceae*	DUF7094 family protein
Q18FD4	*Halobacteriales*	Aryl-sulfate sulfotransferase
Q18FD7	*Halobacteriales*	Phospholipase D-like domain-containing protein
Q18FM1	Only *Haloquadratum*	Ig-halo domain
Q18GI1	*Haloferaceae*	DUF7285 family protein
Q18H44	Only *Haloquadratum*	DUF7537 family protein
Q18H89	*Haloferaceae*	ArnT family glycosyltransferase
Q18HM5	*Halobacteriales*	DUF5787 family protein
Q18HZ6	Only *Haloquadratum*	DUF7519 family protein, PspAB, membrane metallopeptidase operon
Q18IF7	*Halobacteriales* and Gram+	Adenyl succinate synthase, DrmE family protein (DISARM anti-phage system)
Q18IF9	*Halobacteriales*, Gram+ and Gram-	DUF8724 family protein, DrmB family protein
Q18JC6	Only *Haloquadratum*	DUF08073 family protein, exclusive *Halobacteria* domain
Q18JC9	*Haloferaceae*	vWA domain-containing protein
Q18JT2	*Haloferaceae*	DUF5305 family protein

Table 4. Summary of the information obtained from running BLASTn and BLASTp, once restricted to only members of the *Haloquadratum* genus, and once excluding members of the genus.

As can be seen on **Table 4**, the proteins are mostly restricted to the *Haloquadratum* genus, with some of them having homologues outside of the genus, but always inside the *Halobacteria* class. The only exceptions are Q18IF7 and Q18IF9, which have domains that are homologous with the DISARM anti-phage system.

The results for all proteins can be found in the third and fourth sheets of **Supplementary Table ST1**.

As already said before, the proteins that interact with nucleotides are J7RFT8, Q18EI8, Q18F64, Q18IF7 and Q18IF9. J7RFT8 has homology with the *Deinococcus* DR2241 protein family, suggesting a DNA repair function.

Q18IF7 and Q18IF9 are both homologous to different components of the bacterial and archaeal DISARM anti-phage system,[49] thus confirming its existence in this species.

Q18FA8 is exclusive to the *Haloferaceae* family, with present in the genera *Halorubrum* and *Haloplanus*, and it is identified as a member of the DUF7094 family. Q18FD7 is identified as a homologue of the PLD protein family[50] in the order *Halobacteriales*. Q18HM5 is a member of the DUF5787 family in *Halobacteriales*, and Q18HZ6 is an *Haloquadratum* exclusive DUF7519 family protein, with links to the PspAB and a membrane metallopeptidase operon, suggesting that it may be involved in protein transport.[51]

Q18GI1 and Q18H44 are both described as protein-complex associated family proteins, and although they are part of different DUF families (Q18GI1 is a DUF7285 family protein, while Q18H44 is a DUF7537 family), both are exclusive to the *Haloferaceae* family.

Q18JC6 can only be found in *Haloquadratum* and contains the *Halobacteria* exclusive DUF08073 domain. Q18JC9 is a vWA domain-containing protein, vWA domains are common in human and vertebrate extracellular proteins, such as collagen, but intracellular vWA domain-containing proteins are common to all three domains of life.[52]

Q18E01 is a *Haloquadratum* DUF7286 family protein, Q18FD4 can be found all over the *Halobacteriales* order, and has homologies with various aryl-sulfate sulfotransferases, Q18FM1 is exclusive to the genus, and contains an Ig-halo domain, homologous to the immunoglobulins, but exclusive to halophilic archaea. Q18H89 is homologous to various ArnT glycosyltransferases of the *Haloferaceae* family and Q18JT2 is a *Haloferaceae* family DUF5305 family protein.

These entries show that the studied proteins are a mix of known enzyme families and some archaeal-specific understudied families, with some proteins restricted to *Haloquadratum* and others more widely conserved across halophilic archaea. The distinctive *Haloquadratum* proteins are the most relevant ones when trying to understand how *H. walsbyi* ended up colonizing the ecological niches that it occupies.

Clusters of Orthologous Genes

Using a bash script created by Claude Sonnet 4.6, some proteins were able to be classified into different clusters by homology to the list of *H. walsbyi* proteins from the COG database.[43]

UniProtKB code	COG ID	COG Category	COG Description	COG Function
J7RFT8	COG0464	D T M	Cell cycle control, cell division, chromosome partitioning	AAA+-type ATPase, SpoVK/Ycf46/Vps4 family
Q18E01	COG0294	H	Coenzyme transport and metabolism	Dihydropteroate synthase
Q18EI8	COG0644	C	Energy production and conversion	Dehydrogenase (flavoprotein)
Q18FD4	COG0616	O	Post-translational modification, protein turnover, chaperones	Periplasmic serine protease, ClpP class
Q18FM1	COG0618	A T	RNA processing and modification	nanoRNase/pAp phosphatase, hydrolyses c-di-AMP and oligoRNAs
Q18GD8	COG0277	C	Energy production and conversion	FAD/FMN-containing lactate dehydrogenase/glycolate oxidase
Q18GI1	COG0819	H	Coenzyme transport and metabolism	Aminopyrimidine amino hydrolase TenA (thiamine salvage pathway)
Q18H89	COG0461	F	Nucleotide transport and metabolism	Orotate phosphoribosyl transferase
Q18HM5	COG0310	P	Inorganic ion transport and metabolism	ABC-type Co2+ transport system, permease component
Q18IF7	COG1156	C	Energy production and conversion	Archaeal/vacuolar-type H+-ATPase subunit B/Vma2
Q18IK6	COG1213	I	Lipid transport and metabolism	Choline kinase
Q18IW2	COG0394	T	Signal transduction mechanisms	Protein-tyrosine-phosphatase
Q18JC6	COG0619	H	Coenzyme transport and metabolism	ECF-type transporter transmembrane protein EcfT
Q18JT2	COG0447	H	Coenzyme transport and metabolism	1,4-Dihydroxy-2-naphthoyl-CoA synthase

Table 5. Extract of the COG assignment with all of the completely annotated proteins

The complete annotations can be found in **Supplementary Table ST2**.

As seen on **Table 5**, J7RFT8 is assigned to COG0464, which falls in categories D, T, and M (cell cycle control, signal transduction, and cell envelope-related functions), the COG function is an AAA+ type ATPase from the SpoVK/Ycf46/Vps4 family. Q18E01 maps to COG0294 of the H category (coenzyme transport and metabolism), and its function is dihydropteroate synthase. That points to a role in folate biosynthesis, a key pathway for cofactor production. Q18EI8 is assigned COG0644, category C (energy production and conversion), and it is described as a dehydrogenase (flavoprotein). This suggests an oxidoreductase function, likely tied to respiratory or redox metabolism.

Q18FD4 is assigned to COG0616, category O (post-translational modification, protein turnover, and chaperones). Its assigned function is a periplasmic serine protease, ClpP class, indicating a protein quality control or degradation role. Q18FM1 is mapped to COG0618, categories A and T, (RNA processing/modification and signal transduction mechanisms). The function is given as a nanoRNase/pAp phosphatase, with activity on c-di-AMP and oligoRNAs, so this protein likely participates in RNA turnover and nucleotide signaling control.

Q18GD8 belongs to COG0277, category C, and is annotated as a FAD/FMN-containing lactate dehydrogenase/glycolate oxidase. That places it among flavin-dependent redox enzymes, involved in oxidizing small organic acids. Q18GI1 is COG0819, category H, and functions as aminopyrimidine amino hydrolase TenA, part of the thiamine salvage pathway.

Q18H89 is assigned to COG0461, category F (nucleotide transport and metabolism), with the orotate phosphoribosyl transferase function. That means it likely participates in pyrimidine biosynthesis, converting orotate into a nucleotide precursor. Q18HM5 is COG0310, category P (inorganic ion transport and metabolism), and its function is an ABC-type Co2+ transport system. This suggests a membrane transporter involved in cobalt uptake or homeostasis.

Q18IF7 is placed in COG1156, category C, and is described as an archaeal/vacuolar-type H+-ATPase subunit B/Vma2. That indicates a role in ATP-driven proton transport, which is central to energy conservation and ion balance. Q18IK6 is COG1213, category I for (lipid transport and metabolism), with the function of choline kinase. This implies a role in phospholipid metabolism, likely in the synthesis of phosphocholine-containing compounds.

Q18IW2 is predicted to be COG0394, category T, and the function is protein-tyrosine-phosphatase. That means it is likely involved in reversible phosphorylation and regulation of cellular signaling. Q18JC6 is COG0619, category H, with the function ECF-type transporter transmembrane protein EcfT. This points to a membrane component of a transporter system for cofactors or vitamins. Q18JT2 is assigned to COG0447, also category H, and its function is 1,4-dihydroxy-2-naphthoyl-CoA synthase, an enzyme involved in the menaquinone pathway.[53]

KEGG Orthology

Name	KO	Softmax Score	Threshold	Definition
Q18IW2	K11719	0.9517	0.8487	lipopolysaccharide export system protein LptC

Table 6. The only KO assignment that had a Softmax Score higher than its Threshold.

As seen on **Table 6,** the only protein where the KO results were above the internal threshold was Q18IW2, which was assigned K11719 (lipopolysaccharide export system protein LptC) using DeepKOALA.[48] The complete results for the KO, including the annotations below the internal threshold can be seen wholly in **Supplementary Table ST3**.

Discussion

Results Integration

As can be seen on the Results section, **different methods offer significant differences on their predictions**, for example Q18IW2 is an acyltransferase according to I-TASSER,[34] a tyrosine phosphatase according to COG[43] and a lipopolysaccharide export protein according to DeepKOALA.[48] **In the case of a conflict, the I-TASSER prediction will be used** since it is the only one that provides GO codes.

The COG predictions will be used for their categories, not their exact functional prediction.

Even if the I-TASSER predictions are taken at face value, **not all proteins were given specific results**, for example, the only GO term for the MF aspect of protein Q18FM1 was "protein binding", there were 10 different BP routes in which it could take part and "obsolete intracellular part" isn't highly informative, especially if it's the only GO term for CC.

Domains of Unknown Function

Lots of proteins were identified as homologous to various DUF family proteins. **Domains of Unknown Function (DUFs)** are a classification for protein domains that possess a conserved amino acid sequence but have no currently characterized biological function. They are typically named with the prefix "DUF" followed by a number (e.g., DUF5787, DUF11).[54] Despite their lack of annotation, they are ubiquitous across all domains of life; for instance, the InterPro database identifies 4,795 gene families (24%) as DUFs.[54,55]

DUFs represent a **major bottleneck in protein functional annotation** because DUFs by definition have no known function, so **a BLAST hit[21] to a DUF in another species provides no new functional insight**.[56] Many organisms containing unique DUFs, such as marine archaea, thrive under extreme conditions and are difficult to isolate or culture in a laboratory.[56] The difficulty of creating suitable experimental conditions mean there is **very little experimental knowledge** of their tridimensional structure or metabolic roles.[54,56]

The bottleneck increases when **DUFs are systematically overlooked in research** because they are often found in only a few genomes or are perceived as having "little relevance".[54] This neglect persists even though many DUFs are highly conserved, suggesting they perform vital biological roles. DUFs essential role was confirmed with a study of 16 model bacterial species that identified **238 essential DUFs within 355 essential proteins**.[57]

As seen in **Supplementary Table 4**, over 2,700 DUFs are found in prokaryotes, representing a much higher proportion of total domains compared to eukaryotes,[54,55] and **more than half of all known DUFs are detected exclusively in prokaryotes**.[57]

The abundance of DUFs in archaea is mainly due to the extreme ecological niches that they occupy, and it usually results in smaller and structurally more stable proteins than eukaryotes.[56] These unique characteristics and previously undiscovered domain folds are believed to be part of the adaptive strategies used to survive in deep marine environments (high pressure, high salinity, low temperature).[54,56]

As such, **the experimental validation of a DUF's predicted function will single-handedly be able to annotate up to thousands of proteins**.[55]

A protein complex

As seen in **Table 3** and in **Supplementary Table ST1**, proteins Q18H44, Q18IW2 and Q18JC6 all have the BP GO:0046394 "carboxylic acid biosynthetic process" and GO:0006631 "**fatty acid metabolic process**".

When looking at their MF Q18H44, Q18IW2 and Q18JC6 also have some terms in common: GO:0016418 "S-acetyltransferase activity", GO:0016628 "oxidoreductase activity, acting on the CH-CH group of donors, NAD or NADP as acceptor" and GO:0016297 "**fatty acyl-[ACP] hydrolase activity**".

Finally, their CC all have the terms GO:0032991 "**protein-containing complex**" and GO:0043231 "intracellular membrane-bounded organelle". This poses an interesting question, and points to *H. walsbyi* possessing lipid droplets where certain biochemical reactions pertaining to the catabolism of fatty acids take place.[58]

Additional evidence comes from their COG categories, as seen in **Table 5**. Q18GI1 and Q18JC6 are both from the H category (coenzyme transport and metabolism),[43] showing that the reactions that these enzymes catalyse involve coenzyme A. Q18IW2 is from the T category (signal transduction members), which comes as a bit of a surprise since beta oxidation is a completely different pathway to signal transduction.[43]

With the available evidence, it is reasonable to think that these **three proteins form a protein complex inside a lipid droplet and are part of the beta-oxidation of fatty acids**. Trying to be more specific, the "fatty acyl-[ACP] hydrolase activity", the "oxidoreductase activity, acting on the CH-CH group of donors" and the "S-acetyltransferase activity" all point towards the protein complex having a function equivalent to **EC 2.3.1.16**.[13,59]

On the KEGG Pathway (*H. walsbyi* specific) "Fatty acid degradation" page EC 2.3.1.16 is shown to perform a similar function that EC 2.3.1.9 which is annotated as a member of the "Two component system" and explaining the T categorization for Q18IW2.[13,59]

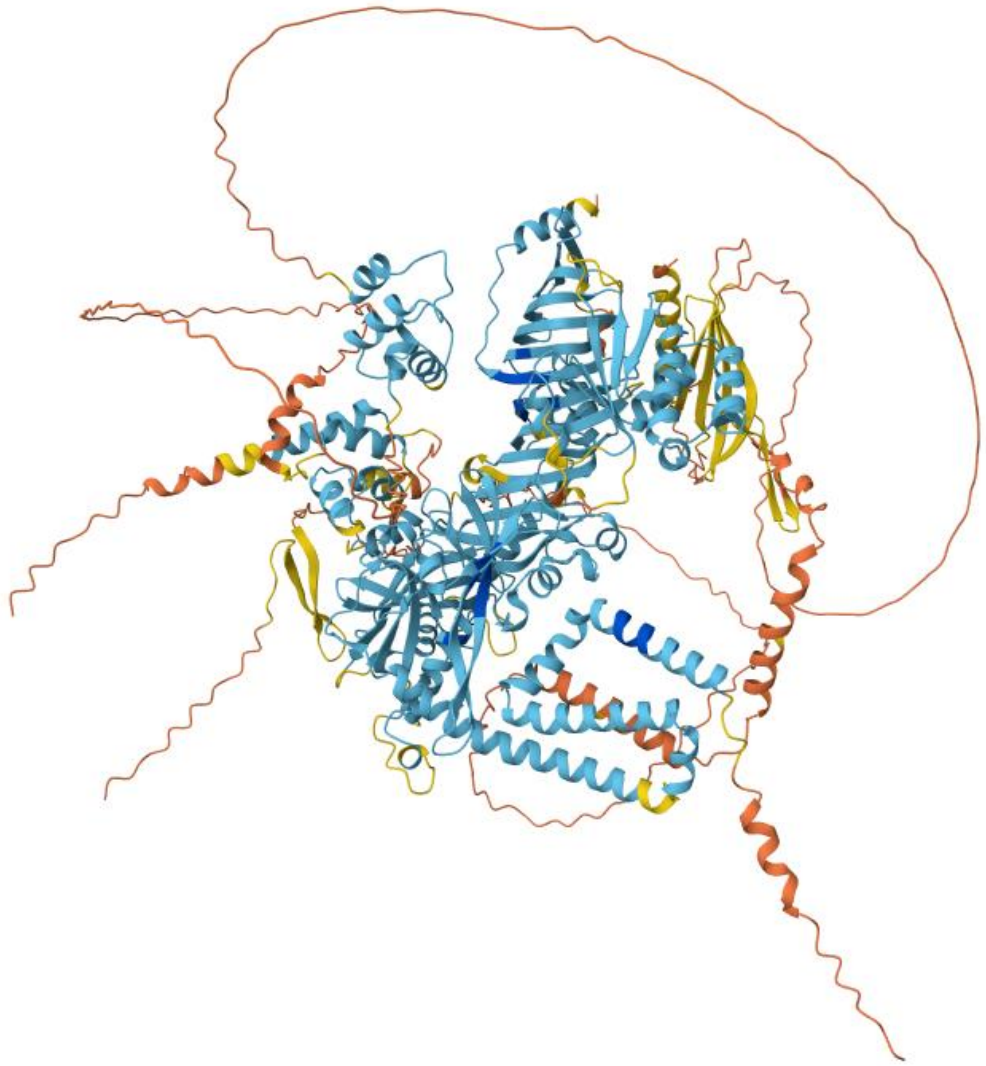

Figure 3. A structural prediction of the putative protein complex that involves Q18GI1, Q18IW2 and Q18JC6. Predicted using AlphaFold3[60] from Google's AlphaFold Server.

Future research

Further **research should be carried out in other understudied organisms**, where the methodology used can be useful in order to identify putative protein functions for a significant percentage of the organism's proteins. This percentage is at its extreme on organisms that aren't the type strain for their species (36% of *Haloquadratum walsbyi* J07HQW2's proteome are uncharacterized proteins),[42] but there still are large amounts of unstudied proteins on type strains such as *Halorubrum salinarum* RHB-C (15%), *Salinibacter ruber* M31 (13%) and *Methanohalophilus halophilus* DSM 3094 (11%).[42]

Moreover, **optimizing the methodology will be necessary**, because more accurate models such as I-TASSER-MTD are significantly slower.[35] This points to a near future with two pipelines, one of fast and lower fidelity predictions of whole proteomes;[34,40,61] and one of more detailed study of individual proteins using specialized models.[18,26,35]

The methodology and specific models won't be the only limiting factor because for some models (such as DeepFRI)[61] the input consists of both sequence and structure data, and if the structure data is an AlphaFold prediction obtained from the unregulated UniProtKB rather than SwissProt, the low quality of some of the AlphaFold structural predictions (there's regions with secondary structure that have a pLDDT lower than fifty) actively hinders the prediction quality from said models.

Finally, the study of the Q18H44, Q18IW2 and Q18JC6 protein complex is not complete, and should be followed up by the annotation of all *H. walsbyi* proteins to find other proteins that might be part of the complex.

Conclusions

This study used GCN and sequence homology models and assigned a putative function to a series of uncharacterized proteins, whose functions will remain putative until experimental data for the individual proteins or their domains can be gathered. Not all proteins were given a putative function, highlighting the difficulty of the task at hand and the need for better models as well as the incompleteness of the common databases.

Expanding the overall knowledge about Domains of Unknown Function and their functions will be the key to increased prediction accuracy in the future, not only because the predictions will be able to be corroborated by homology, but also because some proteins will just have an assigned function via homology.

Finally, this study has found GCNs to be a valid tool for protein function prediction, and as the information pool grows their predictions will eventually be used in the same likeness as metagenome-assembled genomes.

Data Availability

All the data obtained for this project, including the Supplementary Tables and the script used for the COG predictions, can be found in a GitHub repository: https://github.com/ppsm-1810/TFG-ppalmer

Use of AI

This study wouldn't have been possible without AI, since some of the algorithms used for protein function prediction (I-TASSER and DeepKOALA)[34,48] as well as AlphaFold[60] can be classified as AI.

Large Language Models were also used during this study, Google's NotebookLM[62] was used in order to simplify and better understand some of the denser GCN articles. As already mentioned, Anthropic's Claude Sonnet 4.6[45] was used to create a bash script that classified the 26 proteins into COG clusters by homology. Finally, Perplexity's default model[63] was used in order to check for citation errors.

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